

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 03

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



J-K ff

Concept of triggering



Topics to be Covered

Race - Around condition



[Race Around Condition]



- Lets understand what is Race-around condition and when it can happen :

level triggered J-K ff

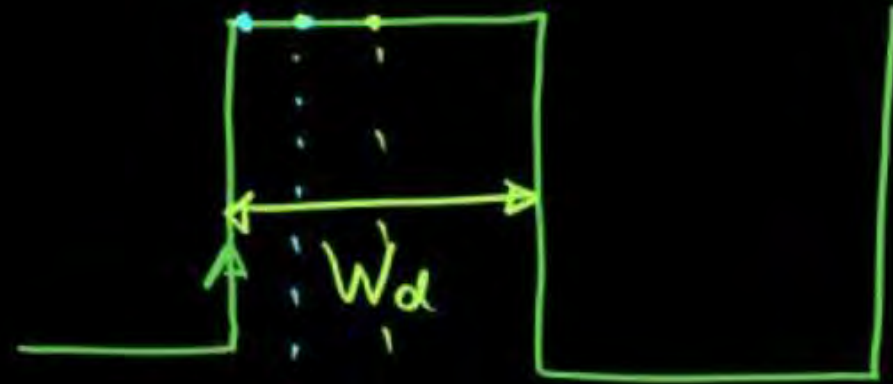
$J=0, K=1$

$Q=1$

$J=1, K=0$

$Q=0$

$J=1, K=1$



$$W_d \gg t_{cf}$$



- In level triggered J-K-ff, when $J=1$ & $K=1$ then O/P toggles many times (more than once) in single clock cycle if $W_d \gg t_{df}$ and this is called as race-around condition.

Race Around condition happens only if :

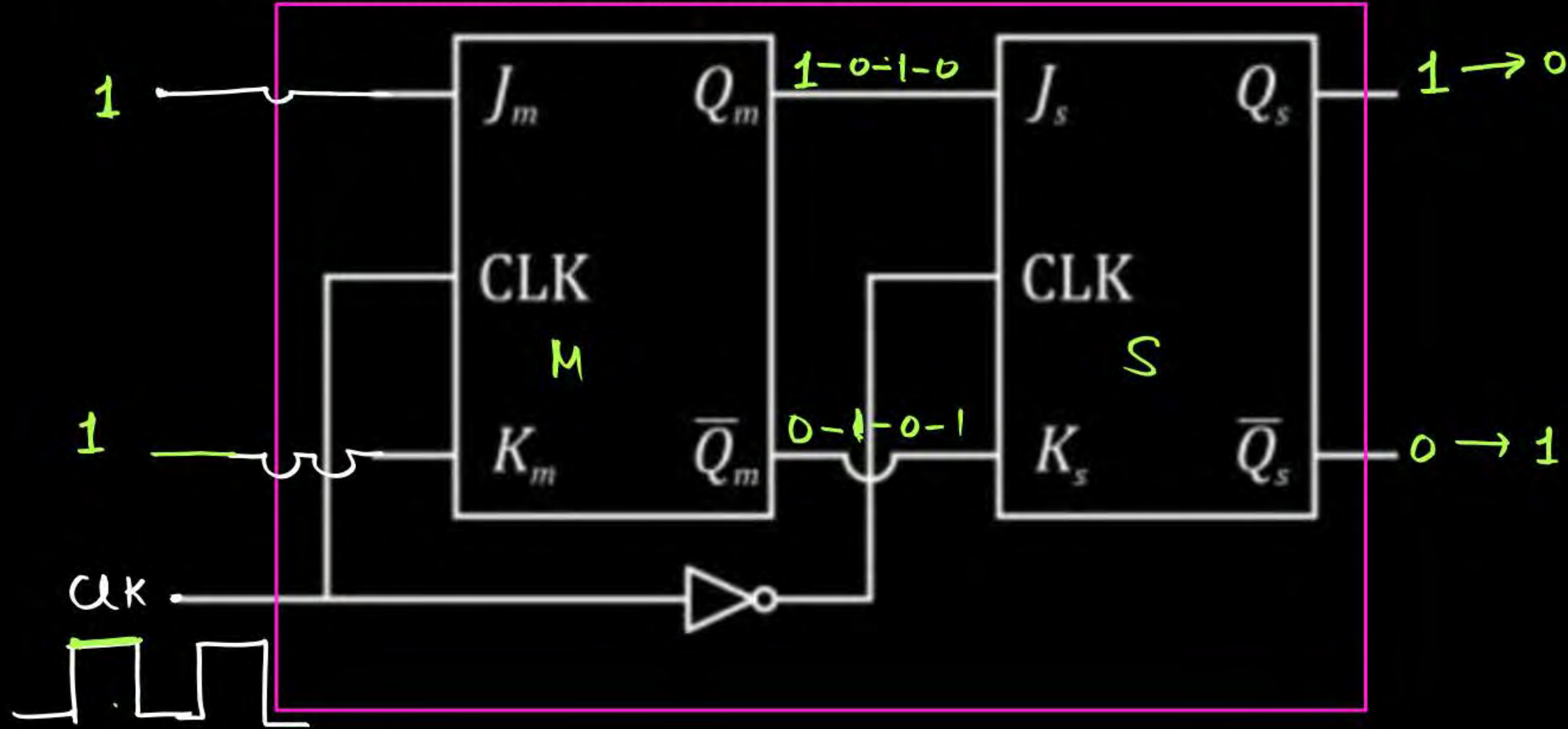
- (i) There must be level triggered J-K ff.
- (ii). J & K must be '1' i.e $J=1$ & $K=1$
- (iii). $W_d \gg t_{df}$.

If all of the above conditions satisfied then only race-around condition happens.

Note: Race-Around condition will never happen in edge triggered J-K ff.

How to avoid race around condition using Master slave FF

- Circuit of master-slave FF :



[How to avoid race around condition using Master slave FF]

- Working :

- In Master-slave ff , O/P is taken from slave - ff.
- In Master-slave ff , i/p is applied to Master ff.
- In Master-slave ff , Master is level triggered while slave is edge triggered.

[**Question**] MSQ



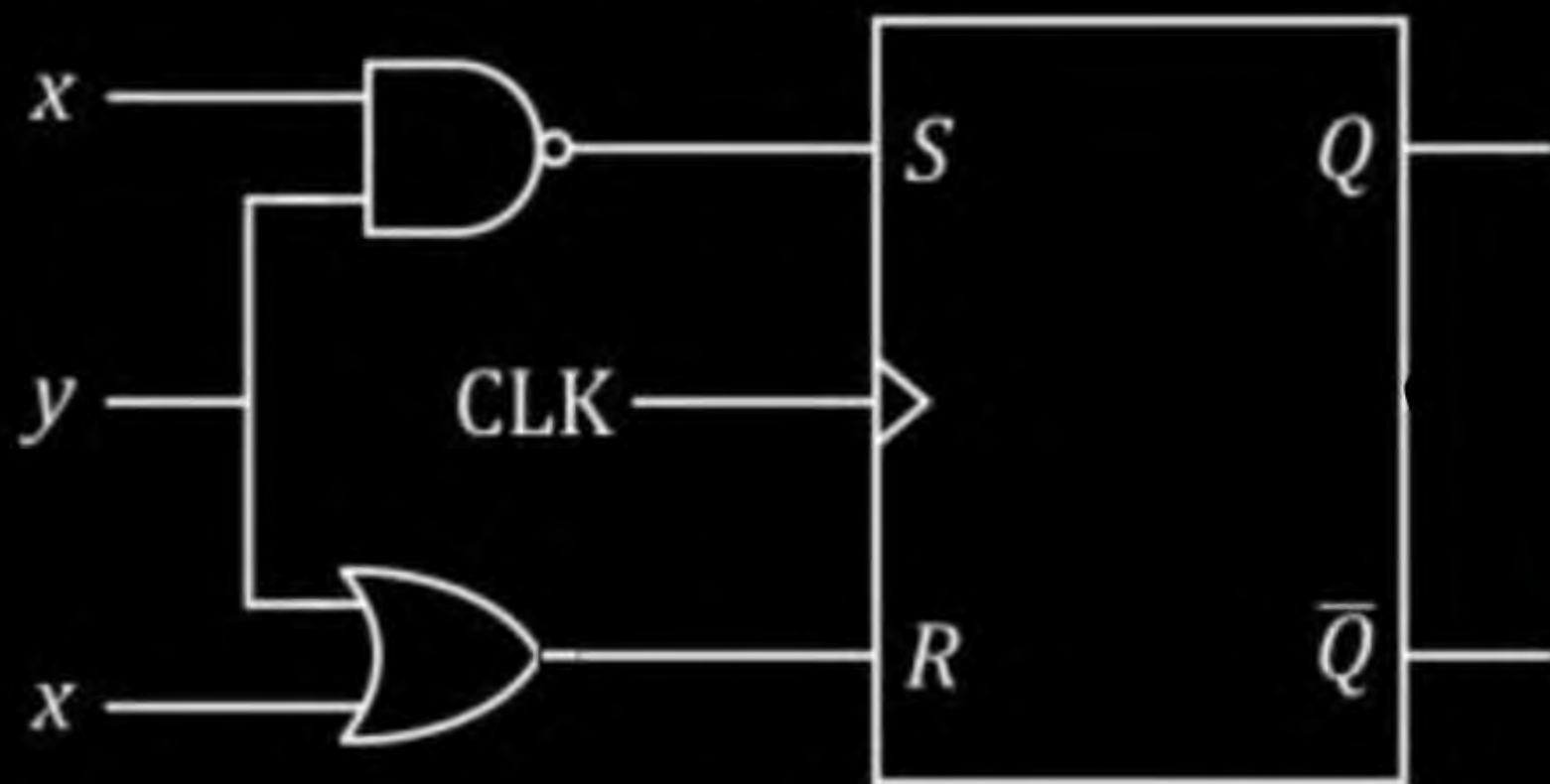
Which of the following is/are true ? (c)

- (a) Race around condition occurs when $J = K = 1$ and edge triggered is applied. ✗
- (b) Race around condition occurs when $J = 1, K = 0$ and level triggered is applied. ✗
- (c) Race around condition never occurs in edge triggered J-K FF. ✓✓
- (d) Race around condition occurs in S-R FF with $S = 0, R = 1$ and level triggered is applied. ✗

[**Question**] MSQ.

An S-R FF circuit is as given below :

ABD



Characteristic equation of above FF will be :

(a) $\bar{x}$

(b) $\bar{y}$

(c) $\overline{x \cdot y}$

(d) $\overline{x + y}$

Valid $S \cdot R = 0$

$$Q(n+1) = S + \bar{R} Q(n)$$

$$= \overline{x \cdot y} + \bar{x} \bar{y} Q(n)$$

$$= \bar{x} + \bar{y} + \bar{x} \bar{y} Q(n)$$

$$= \bar{x} + \bar{y}$$

$$= \overline{x \cdot y} \quad \times$$

$$Q(n+1)[x, y, Q(n)] = \sum 0, 1$$

$$+ d \sum (2, 3, 4, 5)$$

x	y	$Q(n)$	$Q(n+1)$
0	0	0	1
0	0	1	1
0	1	0	X
0	1	1	X
1	0	0	X
1	0	1	X
1	1	0	0
1	1	1	0

	$\bar{y}\bar{Q}$	$\bar{y}Q$	yQ	$y\bar{Q}$
$\bar{x}$	1	1	X	X
x	X	X		

$$Q(n+1) = \bar{y}$$

	$\bar{y}\bar{Q}$	$\bar{y}Q$	yQ	$y\bar{Q}$
$\bar{x}$	1	1	X	X
x	X	X		

$$\rightarrow \bar{x} \quad Q(n+1) = \bar{x}$$

	$\bar{y}\bar{Q}$	$\bar{y}Q$	yQ	$y\bar{Q}$
$\bar{x}$	1	1	X	X
x	X	X		

$$\bar{x}\bar{y} \quad Q(n+1) = \bar{x}\bar{y}$$

[Counters]



- What is counter and for what it is used for ?

- Counter is used for counting no. of clock pulses applied.
- Frequency division

- MOD No. of the counter :

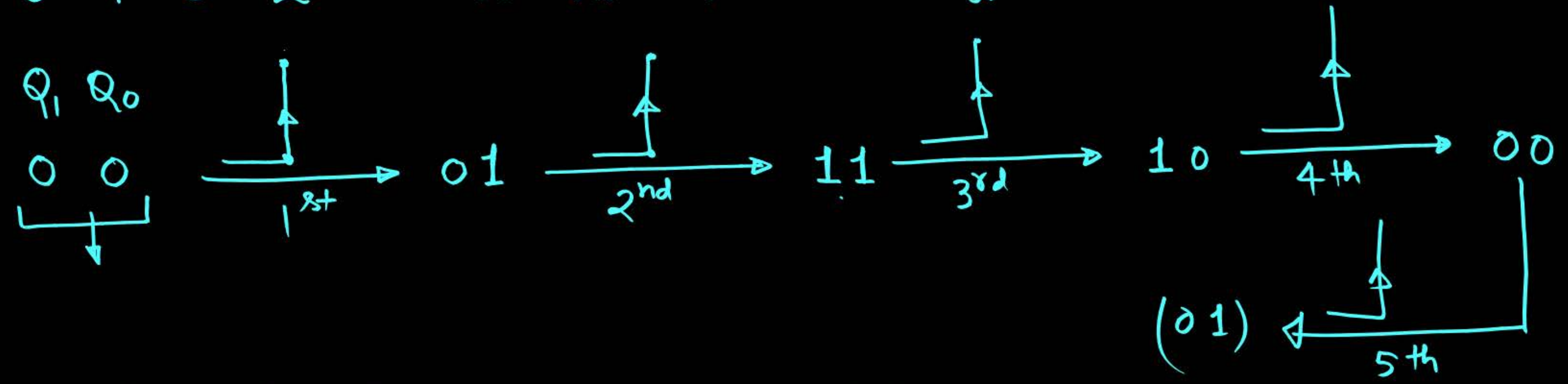
How many different states are there in a counter is called as MOD no. of counter

- $0-1-3-4-2-7$ MOD no. = 6
- $2^0-2^1-2^2-2^3-2^4 \dots 2^m \longrightarrow$ MOD no. = $(m+1)$
- $1-4-9-2-6-8-3-5 \longrightarrow$ MOD no. = 8
- $0-1-2-0-1-3-0-1-4 \longrightarrow$ MOD no. = 9
- $0-1-2-0-1-2-3-0-1-2 \longrightarrow$ MOD no. = 10



• 0-0-1-1-2-2-2-3-3-3-3 $\rightarrow$ MOD no. = 11

• 0-1-3-2 $\rightarrow$ MOD no. = 4 $\rightarrow$ 2ff



- Note: After application of MOD no. of clock pulses counter will again be at the starting state.
- After application of integer multiple of MOD no. of clock pulses counter will again be at starting state.

if starting state is $(3)_{10}$ then after 263 clock pulses, what will be the state of the counter $(1)_{10}$

$$(3)_{10} \xrightarrow{4\text{CLK}} (3)_{10} \xrightarrow{4\text{CLK}} (3)_{10}$$

$$(3)_{10} \xrightarrow[260]{65 \times 4} (3)_{10} \xrightarrow{261} 2 \xrightarrow{262} 0 \xrightarrow{263} 1$$



Topic : 2 Min Summary

- Race-Around condition
- Basics of Counter.

Thank you

GW
Soldiers !

